

Responsible Data Card

Dataset Purpose

This dataset is intended to support the decision of identifying customers who may be at risk of **churning** so that the business can prioritize appropriate customer-retention actions.

The dataset should **not** be used to automatically terminate services, deny customers access to products or services, or make high-impact decisions without appropriate human review.

The prediction should be treated as a **risk signal**, not a definitive statement that a customer will churn.

Provenance and Permission

The dataset is a small customer-level training dataset containing customer identifiers, usage/activity information, spending information, plan type, and historical churn outcomes.

The dataset should only be used for the stated customer-churn analytics purpose and by authorized users. If this dataset represents real customer information, appropriate consent, privacy requirements, and organizational data-use policies must be verified before deployment.

The `customer_id` field should not be used as a predictive feature because it is an identifier rather than a meaningful behavioral attribute.

Population and Representation

The dataset contains **12 customer observations** and represents customers across Basic, Standard, and Pro plans.

Because the dataset is very small, it may not adequately represent the full customer population. The sample may contain limited variation across customer segments, usage patterns, spending levels, and churn behavior.

Therefore, model results from this dataset should not be assumed to generalize to the broader customer population without validation on a larger and representative dataset.

Known limitations include:

- Very small sample size.
- Potentially limited representation of different customer segments.
- Possible class imbalance in the churn target.
- Insufficient data to confidently assess performance across demographic or other sensitive groups.

Features and Target

Feature	Description	Potential Risk
<code>customer_id</code>	Unique customer identifier	Should not be used for prediction
<code>tenure_months</code>	Length of customer relationship in months	May capture lifecycle effects
<code>support_tickets</code>	Number of support tickets raised	Could be associated with dissatisfaction
<code>monthly_spend_inr</code>	Customer's monthly spending in INR	May create socioeconomic/business-segment effects
<code>last_login_days</code>	Number of days since the customer's last login	Strong potential behavioral churn indicator
<code>plan_type</code>	Customer's subscription plan	May act as a proxy for customer segment
<code>churned</code>	Historical churn outcome (0/1)	Prediction target

The target variable is `churned`, where `1` represents a customer who churned and `0` represents a customer who did not churn.

Potential leakage must be checked carefully. Features should only contain information that would have been available **before the prediction decision**. Any information created after churn or after the intervention window must not be included.

No explicit sensitive demographic attributes are present in the dataset. However, features such as spending and plan type could potentially act as proxies for customer characteristics or socioeconomic differences and should therefore be monitored for unintended bias.

Quality Checks

The dataset contains **12 rows and 7 columns**.

Quality checks should include:

- Missing-value analysis for every column.
- Duplicate customer ID checks.
- Validation that numerical values are within reasonable ranges.
- Outlier checks for tenure, support tickets, monthly spending, and inactivity.
- Verification that `churned` contains valid binary values.
- Class-balance analysis between churned and non-churned customers.
- Verification that customer records do not appear in both training and test datasets.
- Confirmation that all predictive features were available before the prediction point.

Because the dataset is extremely small, a simple baseline should be used initially and model performance should not be considered reliable without additional data.

Risks and Safeguards

Bias Risk

The small dataset may not represent the full customer population and could result in biased predictions.

Safeguard: Validate the model on a larger, representative dataset and evaluate performance across relevant customer segments.

Privacy Risk

Customer-level information may contain commercially or personally sensitive information.

Safeguard: Restrict access, minimize personally identifiable information, and use customer identifiers only when operationally necessary.

Misuse Risk

Churn predictions could be incorrectly treated as certain outcomes.

Safeguard: Use predictions as decision-support signals rather than automatic decisions.

False-Positive Risk

A customer predicted as high-risk may actually remain loyal. Unnecessary retention actions could waste resources or negatively affect the customer experience.

Safeguard: Use appropriate probability thresholds and monitor false-positive rates.

False-Negative Risk

A customer predicted as low-risk may actually churn. This could result in a missed retention opportunity.

Safeguard: Monitor recall and false-negative rates and periodically review missed churn cases.

Human Review

Customers receiving significant or potentially intrusive interventions should be subject to appropriate human review rather than fully automated decisions.

Intended Evaluation

Before training a complex model, establish a simple **non-ML baseline**, such as predicting churn based on the historical majority class or a straightforward business rule.

Model evaluation should include:

- **Accuracy** – overall classification correctness.
- **Precision** – proportion of customers predicted to churn who actually churn.
- **Recall** – proportion of actual churners correctly identified.
- **F1-score** – balance between precision and recall.
- **ROC-AUC / PR-AUC** – discrimination performance where appropriate.
- **Calibration** – whether predicted churn probabilities reflect actual churn rates.
- **Confusion matrix** – to explicitly examine false positives and false negatives.
- **Fairness analysis** – compare relevant performance and error rates across customer segments.
- **Error analysis** – investigate examples of both incorrectly predicted churners and missed churners.

Because the current dataset is very small, these evaluation results should be considered **illustrative rather than production-ready**. A larger dataset and an appropriate train/validation/test strategy would be required before deployment.